WORKSHEET 1.7

Where Graphs Break: Classifying Discontinuities

Unit 1: Limits and Continuity

Key ideas

- **Removable:** the limit exists but $f(c)$ is undefined or different. A hole.
- **Jump:** both one-sided limits exist and are unequal.
- **Infinite:** a one-sided limit is $\pm\infty$. A vertical asymptote.
- For a rational function, factor first: a cancelled factor gives a hole; a surviving factor gives an asymptote.

Worked examples

Example 1 · A rational function

Find and classify the discontinuities of $f(x) = \dfrac{x^2 - 4x}{x^2 - 16}$.

Solution. $f(x) = \dfrac{x(x-4)}{(x-4)(x+4)}$. At $x = 4$ the factor cancels and $\lim\limits_{x\to 4} f(x) = \dfrac{4}{8} = \dfrac{1}{2}$. At $x = -4$ the factor remains and the numerator tends to $-4 \neq 0$.

Removable at $x = 4$; infinite at $x = -4$.

Example 2 · A piecewise function

Classify the discontinuity at $x = 2$ of $g(x) = \begin{cases} 2x - 1, & x < 2 \\ 6, & x = 2 \\ x + 1, & x > 2. \end{cases}$

Solution. $\lim\limits_{x\to 2^-} g(x) = 3$ and $\lim\limits_{x\to 2^+} g(x) = 3$, so the limit is 3; but $g(2) = 6$.

Removable.

Example 3 · An absolute value

Classify the discontinuity at $x = 3$ of $h(x) = \dfrac{x^2 - 9}{|x - 3|}$.

Solution. For $x > 3$: $h(x) = x + 3 \to 6$. For $x < 3$: $h(x) = -(x + 3) \to -6$.

Jump.

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Practice

CORE

1. Classify the discontinuity of $f(x) = \dfrac{x^2 - 1}{x - 1}$ at $x = 1$.

2. Classify the discontinuity of $f(x) = \dfrac{3}{x - 2}$ at $x = 2$.

3. Classify the discontinuity at $x = 0$ of $f(x) = \begin{cases} x, & x < 0 \\ x + 2, & x \geq 0. \end{cases}$

4. Classify the discontinuity of $f(x) = \dfrac{x - 5}{(x - 5)^2}$ at $x = 5$.

EXAM LEVEL

5. Find and classify all discontinuities of $f(x) = \dfrac{x^2 + x - 2}{x^2 - x - 6}$.

6. How many vertical asymptotes does the graph of $f(x) = \dfrac{x^2 - 9}{x^3 - 9x}$ have? Justify.

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7. For what value of a does $f(x) = \dfrac{x^2 + ax - 8}{x - 4}$ have a removable discontinuity at $x = 4$? Give the coordinates of the hole.

8. Classify the discontinuity of $f(x) = \dfrac{1 - \cos x}{x}$ at $x = 0$.

Question 9 · Free-response practice

Let $g(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x < 1 \\ k, & x = 1 \\ 3 - x, & x > 1 \end{cases}$ where k is a constant.

(a) Find $\lim\limits_{x \to 1^-} g(x)$ and $\lim\limits_{x \to 1^+} g(x)$.

(b) If $k = 5$, what type of discontinuity does g have at $x = 1$? Give a reason.

(c) For what value of k is g continuous at $x = 1$?

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Worked solutions

1. Removable: $f(x) = x + 1$ for $x \neq 1$, limit 2, $f(1)$ undefined.
2. Infinite: numerator 3, denominator tends to 0.
3. Jump: left limit 0, right limit 2.
4. Infinite: $f(x) = \dfrac{1}{x-5}$ for $x \neq 5$; one factor remains.
5. $f(x) = \dfrac{(x+2)(x-1)}{(x-3)(x+2)}$. Removable at $x = -2$ (limit $\dfrac{-3}{-5} = \dfrac{3}{5}$); infinite at $x = 3$.
6. One. $f(x) = \dfrac{(x-3)(x+3)}{x(x-3)(x+3)} = \dfrac{1}{x}$ for $x \neq \pm 3$: holes at $x = \pm 3$, an asymptote only at $x = 0$.
7. $16 + 4a - 8 = 0$ gives $a = -2$. Then $x^2 - 2x - 8 = (x-4)(x+2)$ and the hole is at $(4, 6)$.
8. Removable: $f(0)$ is undefined but $\lim\limits_{x \to 0} \dfrac{1 - \cos x}{x} = 0$.
9. (a) For $x < 1$, $g(x) = x + 1$, so $\lim\limits_{x \to 1^-} g(x) = 2$. $\lim\limits_{x \to 1^+} g(x) = 3 - 1 = 2$.
   (b) Removable, because $\lim\limits_{x \to 1} g(x) = 2$ exists but $g(1) = 5 \neq 2$.
   (c) $k = 2$, so that $g(1)$ equals the limit.

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